

Revisiting Lattice-based Non-interactive Blind Signature

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Abstract. Blind signatures (BS) allow a signer to produce a valid signature on a message without learning the message itself. They have niche applications in privacy-preserving protocols such as digital cash and electronic voting. Non-interactive blind signatures (NIBS) remove the need for interaction between the signer and the user. In the post-quantum era, lattice-based NIBS schemes are studied as candidates for long-term security.

In ASIACRYPT 2024, Baldimtsi *et al.* proposed the first lattice-based NIBS construction, whose security relies on the random one-more inhomogeneous short integer solution (rOM-ISIS) assumption. This rOM-ISIS is considered to be a non-standard assumption. Later, Zhang *et al.* introduced another lattice-based construction in PROVSEC 2024, and proved its security under the standard module short integer solution (MSIS) assumption. We analyse the security of the latter scheme. In the random oracle model, we show that it fails to achieve both nonce blindness and receiver blindness. We present explicit attacks where an adversary breaks both properties with probability 1. Our attack is based on a crucial observation that uncovers a flaw in the design. Specifically, this flaw allows an attacker to link a message-signature pair with its presignature-nonce pair. In addition, we also identify a flaw in the unforgeability proof. Finally, we suggest a modification to address the issue, which is similar to Baldimtsi *et al.* construction, and its security relies again on the non-standard rOM-ISIS assumption. This work again raises the question of the feasibility of achieving NIBS from standard assumptions.

Keywords: Blind-Signature · Non-interactive BS · Lattice-based NIBS.

1 Introduction

Blind signatures. Blind signatures (BS) [4] is widely used in privacy-preserving systems such as digital currency and electronic voting [5,3,10]. It allows a user to get a signature from a signer without revealing the message being signed. A BS scheme usually requires interaction between the user and the signer. It must satisfy three

main properties. The first is *correctness*, which means that if the signing protocol is executed honestly, the verifier will always accept the message–signature pair. The second is *blindness*, that is, the signer’s view of the protocol does not reveal any information about the blinded message or the final signature. As a result, the signer cannot link the protocol execution to the message–signature pair. The third is *one-more unforgeability* (OMUF), which ensures that a user cannot produce more valid signatures than the number of interactions performed with the signer.

Non-interactive blind signature (NIBS). Lucjan Hanzalik introduced a new primitive called non-interactive blind signature (NIBS) [8]. It is a variant of BS that removes the need for interaction between the signer (issuer) and the user (receiver). In a standard BS protocol, the signer signs a blinded message using its secret key. In contrast, in an NIBS scheme, the receiver holds a secret key and a verification key pair (sk_R, vk_R) . The signer uses its own secret key sk_I to sign the receiver’s verification key vk_R , and produces a pre-signature together with a nonce. From this pre-signature and nonce, the user derives a signature on a random message, resulting in a valid message–signature pair. Finally, the verifier checks this pair using the signer’s verification key vk_I . Figure 1 illustrates the BS and NIBS protocols schematically.

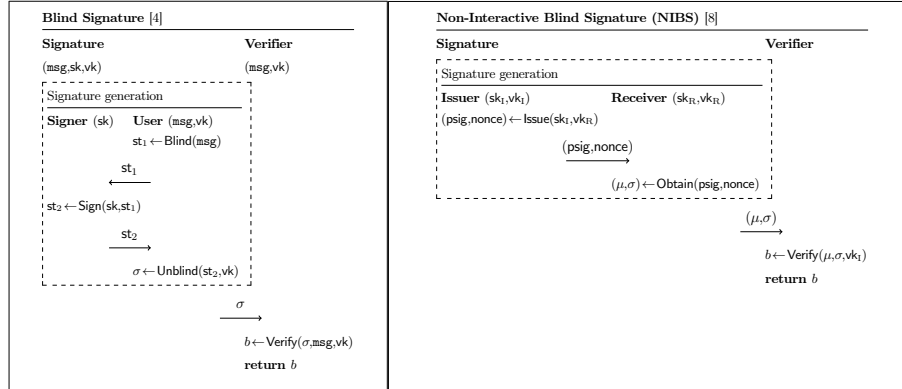


Fig. 1. Schematic diagrams of BS (left) and NIBS (right) protocols.

Fundamental properties of NIBS. For NIBS to be practical, it must provide essential security guarantees. The standard requirements are *correctness*, which is trivial and ensures that honestly generated signatures always verify successfully; *one-more unforgeability*, which ensures that no adversary can generate valid signatures beyond those issued by the signer; and *blindness*, which guarantees unlinkability between presignatures and signatures, irrespective of the nonce used or the receiver’s public key. In the non-interactive setting, blindness becomes more refined and is divided into two forms: *nonce blindness*, which ensures the unlinkability of multiple signatures generated for the same receiver public key under different nonces, and *receiver blindness*, which ensures the unlinkability of signatures issued to different receivers, regardless of the nonce. Together, these definitions address the subtle privacy challenges unique

to NIBS. In addition to these properties, NIBS also requires *reusability*, meaning that a signer can generate multiple distinct pre-signatures for the same receiver’s public key vk_R , with each pre-signature producing a unique message-signature pair.

Applications. NIBS find natural applications in settings such as Privacy Pass, a system currently under discussion in the IETF standardisation process [6]. Privacy Pass aims to reduce the burden of repeated CAPTCHA challenges, particularly for users relying on VPN or TOR. In this system, users obtain tokens from an issuer that can later be redeemed with a content delivery network (CDN) to prove legitimacy without interaction. These tokens play a similar role to electronic coins in Chaum’s original e-cash construction. By replacing traditional blind signatures with NIBS, the interaction between user and issuer is eliminated, while maintaining both blindness and unlinkability, making NIBS particularly well-suited for large-scale, low-latency authentication scenarios. The applications of NIBS are discussed in detail by Hanzalik [8].

Constructions from post-quantum assumptions. Recently, Baldimtsi *et al.* proposed a NIBS construction based on lattice assumptions [2]. Their work strengthens the properties of *reusability* and *blindness*. They adopt the GPV signature scheme [7] from the hash-and-sign paradigm to sign the receiver’s verification key. Later, it uses public-key encryption together with zero-knowledge proofs in the unblinding phase. The security of their construction is proven in the random oracle model (ROM) and relies on a non-standard assumption, namely the random one-more ISIS (rOM-ISIS) assumption. This highlights an interesting open direction: designing a NIBS construction from standard lattice-based assumptions in the ROM. In the same year, Zhang *et al.* [11] proposed another lattice-based NIBS construction. Their scheme employs a verifiable random function (VRF). It uses the GPV signature for presignature generation and incorporates zero-knowledge proofs in the unblinding phase. They claim that their construction achieves security under the module-SIS assumption and provides both forms of blindness, namely *nonce-blindness* and *receiver-blindness*.

Motivation. Designing a secure protocol for privacy-preserving applications such as e-cash or Privacy Pass is the central goal of blind signatures. The primary requirement is to ensure end-to-end security. However, these protocols are inherently complex, as they combine multiple cryptographic components, including hash-and-sign techniques, NIZK proofs, VRFs, commitment schemes, and public-key encryption. Although each building block may be individually secure, integrating them into a single construction can unintentionally leak secret information. As a result, the overall protocol may fail to achieve the desired end-to-end security guarantees.

1.1 Our Contribution.

In this work, we revisit both existing lattice-based NIBS frameworks along with their corresponding security proofs in Section § 3. We carefully scrutinize the construction of [11] and identify several serious security gaps. We observe that the construction of Zhang *et al.* does not achieve blindness, including neither *nonce-blindness* nor *receiver-blindness*, the two key properties of NIBS schemes. Moreover, VRFs are used without a proper analysis of their impact on the security of OMUF. In particular, [11] does not explain how an adversary can extract the necessary witness values from a VRF evaluation without relying on straight-line extractability. We summarize our contributions below.

- In Section §4, we demonstrate that the construction proposed by Zhang *et al.* fails to achieve the fundamental security properties required of a NIBS scheme. In particular, the scheme does not satisfy the blindness property: *nonce blindness* and *receiver blindness* are broken. We show in Theorems 4 and 5 that a malicious adversary can win the corresponding blindness experiments with probability 1. Our analysis assumes that the underlying NIZK proofs are sound, honest-verifier zero-knowledge, and VRF is secure.
- We also observe that the OMUF security proof has a flaw. Since the Zhung *et al.* construction relies on a VRF within the Obtain algorithm. In their OMUF security proof, when the adversary outputs $l+1$ valid message–signature pairs, the reduction is required to extract a witness from one of the signatures. However, the scheme does not employ a straight-line extractable commitment, which forces the reduction to rely on rewinding the adversary. This, in turn, increases the soundness error and weakens the security guarantee. A standard way to address this issue is to incorporate an *extractable commitment* via an *encryption-on-the-sky* mechanism, which allows for straight-line extraction without rewinding.
- In Section §5, we propose a modification to strengthen the construction. In this regard, we adopt a similar approach to [2] and provide a secure version of lattice-based NIBS. As a result, the modified NIBS relies on the rOM-ISIS assumption in the random oracle model, but it is now free from the drawbacks present in [11].

2 Preliminaries

2.1 Notions and Symbols.

Let λ denote the security parameter. We write $\text{negl}(\lambda)$ for a negligible function in λ . Unless stated otherwise, adversaries are assumed to run in probabilistic polynomial time (PPT). We use $\mathcal{R} = \mathbb{Z}[X]/\langle X^d + 1 \rangle$ and $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle X^d + 1 \rangle$ to denote the polynomial rings. For a vector $\mathbf{z} = (z_1, \dots, z_m) \in \mathcal{R}_q^m$, each component satisfies $z_i \in \mathcal{R}_q$. Matrices are defined in boldface. We define a set $\mathbb{S}_\beta$ where the norm of each is less than or equal to β . See Appendix A for basic definitions of cryptographic primitives.

2.2 Lattice Trapdoor.

Trapdoor functions in lattice-based cryptography are constructed using structured matrices sampled from families of hard lattice problems. Let $n, m \in \mathbb{N}$ with $m \geq \mathcal{O}(n \log q)$. A matrix $\mathbf{A} \in \mathcal{R}_q^{n \times m}$, where q is a prime modulus, induces the q -ary lattice

$$\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{y} \in \mathcal{R}_q^m \mid \mathbf{A}\mathbf{y} \equiv \mathbf{0} \pmod{q}\}.$$

A *trapdoor* for A is a short basis $\mathbf{T}_\mathbf{A}$ for $\Lambda_q^\perp(\mathbf{A})$ that enables efficient sampling of short preimages of arbitrary syndromes under the linear map $f_\mathbf{A} : \mathbf{x} \mapsto \mathbf{A}\mathbf{x} \pmod{q}$. We briefly explain how a lattice trapdoor enables the computation of a preimage of a given vector.

- **Parameter generation:** $(q, n, m) \leftarrow \text{Setup}(1^\lambda)$. On input λ , it outputs lattice parameters q, n, m ; where $q \geq 2$, and $m \geq \mathcal{O}(n \log q)$.

- **Trapdoor generation:** $(\mathbf{T}_A, \mathbf{A}) \leftarrow \text{TrapGen}(q, n, m)$. This randomised lattice generation algorithm takes modulus q and the size of the matrix n, m as input and it outputs a matrix $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ and a trapdoor $\mathbf{T}_A$.
- **Preimage sampling:** $\mathbf{s} \leftarrow \text{SamplePre}(\mathbf{T}_A, \mathbf{A}, \mathbf{t})$. It takes the matrix $\mathbf{A}$, trapdoor $\mathbf{T}_A$, a vector $\mathbf{t} \in \mathcal{R}_q^n$ and a parameter $\rho \in \mathbb{R}$. Here, the parameter ρ determines the length of the output vectors. SamplePre outputs a vectors $\mathbf{s} \in \mathcal{R}_q^m$ such that $\mathbf{A}\mathbf{s} = \mathbf{t}$ with $\|\mathbf{s}\| \leq \sqrt{m}\rho$.

Properties. To ensure the hardness, the algorithm must satisfy the following properties.

- **Well-distribution of matrices.** Let us consider the algorithm $(\mathbf{A}, \mathbf{T}_A) \leftarrow \text{TrapGen}(q, n, m)$, which outputs a matrix $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ together with a trapdoor $\mathbf{T}_A$. Let $\mathcal{T}$ denote the distribution of matrices $\mathbf{A}$ generated by TrapGen , and let $\mathcal{U}$ denote the uniform distribution of matrices of dimension $n \times m$ over $\mathcal{R}_q$. Then the distributions $\mathcal{T}$ and $\mathcal{U}$ are statistically indistinguishable, i.e., for every PPT distinguisher $\mathcal{D}$,

$$\left| \Pr[\mathcal{D}(\mathbf{A}) = 1 \mid \mathbf{A} \leftarrow \mathcal{T}] - \Pr[\mathcal{D}(\mathbf{A}) = 1 \mid \mathbf{A} \leftarrow \mathcal{U}] \right| \leq \text{negl}(\lambda).$$

- **Indistinguishability of preimage vector.** Let $(\mathbf{A}, \mathbf{T}_A) \leftarrow \text{TrapGen}(1^n, 1^m, q)$, where $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ and $\mathbf{T}_A$ is its trapdoor. For any $\mathbf{t} \in \mathcal{R}_q^n$ and parameter ρ , compute $\mathbf{s} \leftarrow \text{SamplePre}(\mathbf{A}, \mathbf{T}_A, \mathbf{t}, \rho)$, with $\|\mathbf{s}\| \leq \rho$. Let $\mathcal{S}_\rho$ be the distribution of such $\mathbf{s}$, and $\mathcal{G}_{\mathcal{R}_q^m, \rho}^{\mathbb{S}}$ the discrete Gaussian distribution. Then for every PPT distinguisher $\mathcal{D}$,

$$\left| \Pr[\mathcal{D}(\mathbf{s}) = 1 \mid \mathbf{s} \leftarrow \mathcal{S}_\rho] - \Pr[\mathcal{D}(\mathbf{s}) = 1 \mid \mathbf{s} \leftarrow \mathcal{G}_{\mathcal{R}_q^m, \rho}^{\mathbb{S}}] \right| \leq \text{negl}(\lambda).$$

2.3 Non-interactive Blind Signature.

Non-interactive blind signatures (NIBS) were first introduced by Lucjan Hanzalik, along with their tagged version (TNIBS) [8]. We present the definition of NIBS below. In the following definition, the differences specific to TNIBS are marked in red. Later, we present the security properties for NIBS.

Definition 1 (Tagged Non-Interactive Blind Signature). A (tagged) non-interactive blind signature $\text{TNIBS} = (\text{KeyGen}, \text{RecKeyGen}, \text{Issue}, \text{Obtain}, \text{Verify})$ has five different algorithms.

- $(\text{sk}_I, \text{vk}_I) \leftarrow \text{KeyGen}(1^\lambda)$: The key generation algorithm takes the security parameter λ as input and outputs the issuer's key pair $(\text{sk}_I, \text{vk}_I)$. This key pair also serves as the key pair for the NIBS scheme.
- $(\text{sk}_R, \text{vk}_R) \leftarrow \text{RecKeyGen}(1^\lambda)$: The key generation algorithm takes λ as input, and it outputs a key pair for the $\text{rec}(\text{sk}_R, \text{vk}_R)$.
- $(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_I, \text{vk}_R, \tau)$: The signature generation algorithm takes issuer's secret key sk_I , and receiver's verification key vk_R (and tag τ) as input, and it outputs $(\text{psig}, \text{nonce})$.
- $(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_R, \text{vk}_I, \text{psig}, \text{nonce}, \tau)$: On input a user secret key sk_R , a issuer public key vk_I , a pre-signature psig , and a randomness nonce , (and tag τ) outputs a message-signature pair (μ, σ) or $\perp$.

– $b \leftarrow \text{Verify}(\text{vk}_I, \mu, \sigma, \tau)$: It takes public key vk_I , and a message-signature pair (μ, σ) , (and tag τ) as input, deterministically outputs a bit b . If $b=1$, accept the signature σ ; otherwise, reject it.

Definition 2 (Correctness). A TNIBS scheme is correct if, for all security parameters λ , for all key pairs $(\text{sk}_I, \text{vk}_I) \leftarrow \text{KeyGen}(1^\lambda)$, $(\text{sk}_R, \text{vk}_R) \leftarrow \text{RecKeyGen}(1^\lambda)$ and for all messages μ , the following holds. If $(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_I, \text{vk}_R, \tau)$, and $(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_R, \text{vk}_I, \text{psig}, \text{nonce}, \tau)$, then

$$\text{Verify}(\text{vk}_I, \mu, \sigma, \tau) = 1$$

with probability 1 over the randomness of all algorithms.

We only present the OMUF and blindness properties for NIBS. The analogous properties for TNIBS can be established in a similar manner.

Definition 3 (One-More Unforgeability of NIBS). Let $\mathcal{A}$ be a PPT adversary against the OMUF security of a NIBS scheme NIBS. The advantage of $\mathcal{A}$ in the experiment $\text{Exp}_{\text{NIBS}}^{\text{OMUF}}(\mathcal{A}, 1^\lambda)$ (see Figure: 2) is

$$\text{Adv}_{\text{NIBS}}^{\text{OMUF}}(\mathcal{A}) = \Pr[\text{Exp}_{\text{NIBS}}^{\text{OMUF}}(\mathcal{A}, 1^\lambda) = 1].$$

We say that NIBS is one-more unforgeable if $\text{Adv}_{\text{NIBS}}^{\text{OMUF}}(\mathcal{A})$ is negligible for all PPT adversaries $\mathcal{A}$.

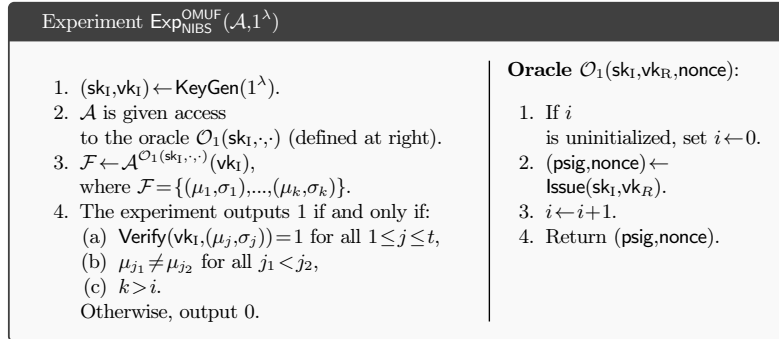


Fig. 2. One-more unforgeability for NIBS

Definition 4 (Receiver Blindness). Let $\mathcal{A}$ be a PPT adversary against the receiver blindness of a NIBS scheme. We define the advantage of $\mathcal{A}$ in the receiver blindness experiment $\text{Exp}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda)$ as:

$$\text{Adv}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}) = \left| \Pr[\text{Exp}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda) = 1] - \frac{1}{2} \right|.$$

The NIBS scheme satisfies receiver blindness if $\text{Adv}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A})$ is negligible in the security parameter λ for all PPT adversaries $\mathcal{A}$.

Definition 5 (Nonce Blindness). Let $\mathcal{A}$ be a PPT adversary against the nonce-blindness security of a NIBS scheme. We say that NIBS satisfies nonce blindness if, for all such adversaries $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that

$$\text{Adv}_{\text{NIBS}}^{\text{NBLD}}(\mathcal{A}) \leq \text{negl}(\lambda),$$

where $\text{Adv}_{\text{NIBS}}^{\text{NBLD}}(\mathcal{A}) = \left| \Pr[\text{Exp}_{\text{NIBS}}^{\text{NBLD}}(\mathcal{A}, 1^\lambda) = 1] - \frac{1}{2} \right|$, and $\text{Exp}_{\text{NIBS}}^{\text{NBLD}}$ is the nonce-blindness experiment.

Definition 6. A NIBS scheme is said to be fully blind if it satisfies both receiver blindness and nonce blindness (see Figure 3), i.e., for every PPT adversary $\mathcal{A}$,

$$\text{Adv}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}) \approx 0 \quad \text{and} \quad \text{Adv}_{\text{NIBS}}^{\text{NBLD}}(\mathcal{A}) \approx 0.$$

Blindness Experiments for NIBS	
Nonce Blindness: $\text{Exp}_{\text{NIBS}}^{\text{NBLD}}(\mathcal{A}, 1^\lambda)$ 1. $(\text{sk}_R, \text{vk}_R) \xleftarrow{\\$} \text{KeyGen}(1^\lambda)$. 2. $(\text{psig}_0, \text{nonce}_0, \text{psig}_1, \text{nonce}_1, \text{vk}_I) \leftarrow \mathcal{A}(\text{vk}_R)$. 3. $(\mu_i, \sigma_i) \leftarrow \text{Obtain}(\text{sk}_{R_i}, \text{vk}_I, \text{psig}_i, \text{nonce}_i)$ for $i \in \{0, 1\}$ 4. If $\sigma_0 = \perp$ or $\sigma_1 = \perp$, then set $(\mu_0, \sigma_0) := \perp$ and $(\mu_1, \sigma_1) := \perp$. 5. Choose $b \xleftarrow{\\$} \{0, 1\}$. 6. $b_1 \leftarrow \mathcal{A}((\mu_b, \sigma_b), (\mu_{1-b}, \sigma_{1-b}))$. 7. Output 1 iff $b_1 = b$.	Receiver Blindness: $\text{Exp}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda)$ 1. $(\text{sk}_{R_i}, \text{vk}_{R_i}) \xleftarrow{\\$} \text{RecKeyGen}(1^\lambda)$ for $i \in \{0, 1\}$. 2. $(\text{psig}_0, \text{nonce}_0, \text{psig}_1, \text{nonce}_1, \text{vk}_I) \leftarrow \mathcal{A}(\text{vk}_{R_0}, \text{vk}_{R_1})$. 3. $(\mu_i, \sigma_i) \leftarrow \text{Obtain}(\text{sk}_{R_i}, \text{vk}_I, \text{psig}_i, \text{nonce}_i)$ for $i \in \{0, 1\}$ 4. If $\sigma_0 = \perp$ or $\sigma_1 = \perp$, then set $(\mu_0, \sigma_0) := \perp$ and $(\mu_1, \sigma_1) := \perp$. 5. Choose $b \xleftarrow{\\$} \{0, 1\}$. 6. $b_1 \leftarrow \mathcal{A}((\mu_b, \sigma_b), (\mu_{1-b}, \sigma_{1-b}))$. 7. Output 1 iff $b_1 = b$.

Fig. 3. Blindness property of NIBS

Another fundamental property of NIBS is *reusability*. Informally, this requires that a receiver should be able, except with negligible advantage $\text{negl}(\lambda)$, to obtain two distinct signed messages $\mu_0 \neq \mu_1$ (along with their valid signatures), corresponding to two independently generated nonce-presignature pairs, while using the same verification key vk_R .

Definition 7 (Reusability of NIBS). A NIBS scheme satisfies the reusability property if there exists a negligible function $\text{negl}(\lambda)$ such that for every $\lambda \in \mathbb{N}$, the advantage that the experiment $\text{Exp}_{\text{NIBS}}^{\text{REUSE}}(\mathcal{A}, 1^\lambda)$ outputs 1 is at most $\text{negl}(\lambda)$.

The reusability advantage of a receiver is then defined as

$$\text{Adv}_{\text{NIBS}}^{\text{REUSE}}(\mathcal{A}) = \Pr[\text{Exp}_{\text{NIBS}}^{\text{REUSE}}(\mathcal{A}, 1^\lambda) = 1].$$

A NIBS scheme is *reusable* if $\text{Adv}_{\text{NIBS}}^{\text{REUSE}}(\mathcal{A}) \leq \text{negl}(\lambda)$.

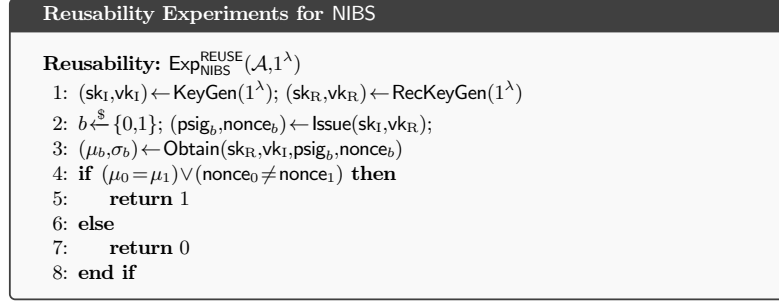


Fig. 4. Reusability property of NIBS

3 Revisiting Lattice-based NIBS from Module-SIS.

In EUROCRYPT 2023, Hanzlik [8] introduced the notion of non-interactive blind signatures (NIBS), and left open the problem of constructing such primitives in the post-quantum setting. Subsequently, two works appeared in the literature addressing this question. In ASIACRYPT 2024, Baldimtsi *et al.* [2] proposed a lattice-based NIBS construction relying on the non-standard assumption rOM-ISIS. In PROVSEC 2024, Zhang *et al.* [11] presented another lattice-based NIBS construction, which they showed can be based on the standard MSIS problem. For brevity, we refer to the constructions of Baldimtsi *et al.* [2] as BCGY (presented in Figure: 7, Appendix B), and those of Zhang *et al.* [11] as ZCH. In the following, we describe the ZCH NIBS construction.

3.1 ZCH Lattice-based NIBS

Let ZCH.LNIBS consists of five different polynomial-time algorithms: $\{\text{KeyGen}, \text{RecKeyGen}, \text{Issue}, \text{Obtain}, \text{Verify}\}$.

– **Issuer Key Generation:** $(\text{sk}_I, \text{vk}_I) \leftarrow \text{KeyGen}(1^\lambda)$. The issuer runs $\text{SetUp}(1^\lambda)$ to obtain public parameters params and then executes $\text{TrapGen}(\text{params})$ to generate a matrix-trapdoor pair $(\mathbf{A}, \mathbf{T}_\mathbf{A})$. It sets its secret key to $\text{sk}_I = \mathbf{T}_\mathbf{A}$ and its verification key to $\text{vk}_I = \mathbf{A}$.

– **Receiver Key Generation:** $(\text{sk}_R, \text{vk}_R) \leftarrow \text{RecKeyGen}(1^\lambda)$. The receiver samples a matrix $\mathbf{B} \xleftarrow{\$} \mathcal{R}_q^{m \times n_1}$ with $n_1 > m$ and a secret vector $\mathbf{s} \xleftarrow{\$} \mathbb{S}_\beta^{n_1}$, then computes $\mathbf{u} = \mathbf{B}\mathbf{s} \in \mathcal{R}_q^n$. It sets its secret key to $\text{sk}_R = \mathbf{s}$ and its verification key to $\text{vk}_R = (\mathbf{u}, \mathbf{B})$.

– **Issue Algorithm:** $(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_I, \text{vk}_R)$. Given $(\text{sk}_I, \text{vk}_R)$, the issuer samples a nonce $\text{nonce} \xleftarrow{\$} \mathcal{R}_q^m$ and computes $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$. Using the lattice trapdoor $\text{sk}_I = \mathbf{T}_\mathbf{A}$, it runs $\text{SamplePre}(\mathbf{A}, \mathbf{T}_\mathbf{A}, \mathbf{t}, \rho)$ to obtain a short vector $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{t}$ and $\|\mathbf{x}\| \leq \beta$. It sets $\text{psig} = \mathbf{x}$ and outputs $(\text{psig}, \text{nonce})$.

– **Obtain Algorithm:** $(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_R, \text{vk}_I, \text{psig}, \text{nonce})$. On input the presignature and nonce pair $(\text{psig}, \text{nonce})$, the receiver reconstructs $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$ and verifies that $\mathbf{A}\mathbf{x} = \mathbf{t}$ and $\|\mathbf{x}\| \leq \beta$ for $\mathbf{x} = \text{psig}$; if the check fails it outputs $\perp$. Otherwise, it computes $(\mu, \pi_{\text{VRF}}) \leftarrow \text{VRF.Eval}(\mathbf{A}, \text{pk}_R, \text{sk}_R, \text{nonce})$ and produces a NIZK

proof π attesting that $\text{VRF.Verify}(\text{vk}_R, \pi_{\text{VRF}}, \text{nonce}, \mu) = 1$, $\mathbf{Ax} = \mathbf{t}$, and $\|\mathbf{x}\| \leq \beta$ hold simultaneously. It sets the signature $\sigma = \pi$ and outputs (μ, σ) .

Construction of NIZK proof π . The proof π consists of two NIZK proofs, denoted π_{LTC} and π_{VRF} . For clarity, the underlying languages for these proofs are stated explicitly. The language associated with π_{LTC} is denoted $\mathcal{L}_{\text{LTC}}$ and is defined as

$$\mathcal{L}_{\text{LTC}} = \{(\mathbf{x}; \mathbf{t}, \mathbf{A}) : \mathbf{Ax} = \mathbf{t} \wedge \|\mathbf{x}\| \leq \beta\}.$$

In $\mathcal{L}_{\text{LTC}}$, the statement for the NIZK proof π_{LTC} is $\text{stm}_{\text{LTC}} = (\mathbf{t}, \mathbf{A})$, and the witness is $\text{wt}_{\text{LTC}} = \mathbf{x}$. The proof π_{LTC} is constructed using the proof techniques described by Lyubashevsky *et al.* [9].

The language corresponding to π_{VRF} is denoted $\mathcal{L}_{\text{VRF}}$. In this case, the statement is $\text{stm}_{\text{VRF}} = (\mu, \text{vk}_I)$, and the witness is $\text{wt}_{\text{VRF}} = (\text{nonce}, \mathbf{x}, \text{sk}_R, \text{vk}_R)$. The language is defined as

$$\mathcal{L}_{\text{VRF}} = \{(\text{wt}_{\text{VRF}}, \text{stm}_{\text{VRF}}) : \text{VRF.Verify}(\text{vk}_R, \pi_{\text{VRF}}, \text{nonce}, \mu) = 1\}.$$

– **Verification Algorithm:** $b \leftarrow \text{Verify}(\pi, \text{vk}_I)$. The verifier takes as input the message μ , the signature σ , and the public parameters **params**. It parses σ to obtain the proof π and verifies it using the NIZK verification procedure. The verifier accepts if and only if the verification of π is successful, that is, if both NIZK proofs π_{LTC} and π_{VRF} are verified successfully.

Tagged ZCH.LNIBS. The tagged NIBS scheme is similar to the NIBS (as defined in the Definition 1). In **Issue** algorithm, the issuer takes a tag τ as an extra input and computes $\mathbf{t} \leftarrow \mathcal{H}(\text{nonce} \parallel \text{vk}_R \parallel \tau)$. The issuer sends the tag (along with presignature and nonce pair) to the receiver. The receiver uses it in **Obtain** to compute $\mathbf{t}$. We refer to the original paper for more details. In what follows, we state the security properties only for NIBS, since TNIBS satisfies the same properties.

3.2 Security Properties of ZCH.LNIBS

For clarity and formal consistency, we restate the main theorems from [11] corresponding to each security property of the ZCH scheme. In each case, the statement is given in a self-contained manner, using the notation adopted in this work. Proofs follow directly from the arguments presented in [11] and are omitted here.

Theorem 1 (One-More Unforgeability). *If the NIZK proofs are sound, the VRF is unique, and the hash function is modelled as a random oracle, then the scheme is unforgeable assuming the hardness of the MSIS problem. That is, no efficient adversary can create more valid signatures than those it legitimately obtained from the signer, except with negligible probability.*

Theorem 2 (Nonce Blindness). *Assume the NIZK proof system is honest-verifier zero knowledge and the VRF is pseudorandom. Then, for every PPT adversary $\mathcal{A}$, the nonce-blindness advantage in the experiment $\text{Exp}_{\text{ZCH.LNIBS}}^{\text{NBLD}}(\mathcal{A}, 1^\lambda)$ is negligible in λ ; that is, $\text{Adv}_{\text{ZCH.LNIBS}}^{\text{NBLD}}(\mathcal{A}) \leq \text{negl}(\lambda)$.*

Lattice-based NIBS	
Key Generation.	
Issuer	Receiver
<u>KeyGen(1^λ)</u>	<u>RecKeyGen(1^λ)</u>
1. $\text{params} \leftarrow \text{SetUp}(1^\lambda)$. 2. $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \leftarrow \text{TrapGen}(\text{params})$. 3. $\text{sk}_\mathbf{I} \leftarrow \mathbf{T}_\mathbf{A}; \text{vk}_\mathbf{I} \leftarrow \mathbf{A}$. 4. return $(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{I})$.	1. $\mathbf{B} \xleftarrow{\\$} \mathcal{R}_q^{m \times n_1}$, where $n_1 > m$. 2. $\mathbf{s} \xleftarrow{\\$} \mathcal{S}^{n_1}$. 3. $\mathbf{u} \leftarrow \mathbf{B}\mathbf{s}; \mathbf{u} \in \mathcal{R}_q^n$. 4. $\text{sk}_\mathbf{R} \leftarrow \mathbf{s}; \text{vk}_\mathbf{R} \leftarrow (\mathbf{u}, \mathbf{B})$. 5. return $(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{R})$.
Signature Generation.	
<u>$(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{R})$</u>	<u>$(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{I}, \text{psig}, \text{nonce})$</u>
1. $\text{nonce} \xleftarrow{\\$} \mathcal{R}_q^m$. 2. $\mathbf{t} \leftarrow \mathcal{H}(\text{nonce} \parallel \text{vk}_\mathbf{R})$. 3. $\mathbf{x} \xleftarrow{\\$} \text{SamplePre}(\mathbf{A}, \mathbf{T}_\mathbf{A}, \mathbf{t}, \rho)$ such that $\mathbf{A}\mathbf{x} = \mathbf{t}$ and $\ \mathbf{x}\ \leq \beta$. 4. $\text{psig} \leftarrow \mathbf{x}$. 5. return $(\text{psig}, \text{nonce})$.	1. Verify $\mathbf{A}\mathbf{x} = \mathbf{t} \wedge \ \mathbf{x}\ \leq \beta$. If it fails, output $\perp$ and abort. 2. $(\mu, \pi_{\text{VRF}}) \leftarrow \text{VRF.Eval}(\mathbf{A}, \text{pk}_\mathbf{R}, \text{sk}_\mathbf{R}, \text{nonce})$. 3. Generate a NIZK proof π for the relation: Given $\text{st} := (\mu, \text{vk}_\mathbf{I})$, there exists $\text{wt} = (\text{nonce}, \text{psig}, \text{vk}_\mathbf{R})$ such that: $\text{VRF.Verify}(\text{vk}_\mathbf{R}, \pi_{\text{VRF}}, \text{nonce}, \mu) = 1$ $\wedge \{\mathbf{A}\mathbf{x} = \mathbf{t} \wedge \ \mathbf{x}\ \leq \beta\}.$ 4. $\sigma \leftarrow \pi$. 5. return (μ, σ).

Fig. 5. ZCH Lattice based non-interactive blind signature [11]

Theorem 3 (Receiver Blindness). *Suppose the NIZK proof system satisfies honest-verifier zero-knowledge and the VRF offers pseudorandomness, then, for every PPT adversary $\mathcal{A}$, the receiver-blindness advantage in the experiment $\text{Exp}_{\text{NIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda)$ is negligible in λ ; that is, $\text{Adv}_{\text{ZCH.LNIBS}}^{\text{RBLD}}(\mathcal{A}) \leq \text{negl}(\lambda)$.*

The authors of the ZCH construction argued that their scheme does not require any non-standard assumptions in the OMUF security proof. They presented a reduction of the OMUF security of ZCH.LNIBS to the Module-SIS problem. However, we observe that ZCH.LNIBS does not provide the blindness property. Breaking blindness requires no specific algorithm. The adversary can simply observe the statement variables of the NIZK proof and thereby establish a link between the presignature–nonce pair with the message–signature pair.

4 Breaking the Blindness of ZCH.LNIBS

The blindness property of NIBS provides unlinkability between a presignature and the final signature from the signer’s view. This unlinkability holds regardless of which

public key or nonce is used. To prove blindness, the scheme must satisfy both nonce blindness and receiver blindness. Nonce blindness considers the case where a malicious signer tries to link a presignature with its corresponding signature. It ensures that multiple signatures under the same public key remain unlinkable. Receiver blindness, on the other hand, ensures that signatures issued to different public keys are unlinkable, independent of the nonce.

Attack idea. The nonce blindness attack exploits that the signing nonce nonce is embedded directly into the binding value $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$. We observe that the value $\mathbf{t}$ appears in the statement of the proof π_{LTC} . So, the adversary can easily link it back to the chosen nonce with certainty. Thus, it breaks nonce blindness. Likewise, the receiver blindness attack arises because $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$ depends on the receiver's verification key vk_R . By comparing the $\mathbf{t}$ values in signatures with those computed from issuance transcripts, the adversary identifies the target receiver with probability one, thereby breaking receiver blindness. Since the scheme does not satisfy nonce blindness or receiver blindness, it is not fully blind. The same weakness applies to the TNIBS construction.

Theorem 4 (Breaking the Nonce-Blindness). *In the random oracle model, the ZCH.LNIBS construction does not satisfy nonce blindness. More precisely, there exists an adversary $\mathcal{A}$ such that*

$$\text{Adv}_{\text{ZCH.LNIBS}}^{\text{NBLD}}(\mathcal{A}) = \Pr \left[\text{Exp}_{\text{ZCH.LNIBS}}^{\text{NBLD}}(\mathcal{A}, 1^\lambda) = 1 \right] - \frac{1}{2} = 1,$$

where $\mathcal{A}$ behaves as follows.

- Precomputation. Obtain two valid issue transcripts $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_R)$ for $i \in \{0, 1\}$. For each i , compute

$$\mathbf{t}_i \leftarrow \mathcal{H}(\text{nonce}_i \parallel \text{vk}_R).$$

- Challenge. Receive the two final outputs $((\mu_0, \sigma_0), (\mu_1, \sigma_1))$, where each σ_i contains a NIZK proof over the public statement $(\mathbf{t}, \mathbf{A})$. From each σ_i , extract the embedded statement value $\mathbf{t}_i^{(\text{sig})}$.
- Decision. Output b such that $\mathbf{t}_b^{(\text{sig})} = \mathbf{t}_j$ for some $j \in \{0, 1\}$, and return the corresponding index j . This ensures that $\Pr[\text{Exp}_{\text{ZCH.LNIBS}}^{\text{NBLD}}(\mathcal{A}, 1^\lambda) = 1] = 1$.

Proof. In the nonce-blindness experiment, the adversary observes two issuance transcripts $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_R)$ for $i \in \{0, 1\}$ before receiving the two challenge signatures. For each transcript, the adversary computes $\mathbf{t}_i := \mathcal{H}(\text{nonce}_i \parallel \text{vk}_R)$ by querying the random oracle. Each challenge signature contains a NIZK whose public statement reveals a value $\mathbf{t}^{(\text{sig})}$ that the verifier checks against the relation $\mathbf{Ax} = \mathbf{t}$ together with $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$. By completeness of the NIZK, any accepted signature must satisfy these equalities. Therefore the value $\mathbf{t}^{(\text{sig})}$ appearing in the accepted signature equals $\mathcal{H}(\text{nonce}_j \parallel \text{vk}_R)$ for the unique $j \in \{0, 1\}$ corresponding to the issuance pair that produced the signature. The adversary compares $\mathbf{t}^{(\text{sig})}$ with $\{\mathbf{t}_0, \mathbf{t}_1\}$ and recovers the correct index with probability 1 over the coins of the experiment and the random oracle. This linkage does not rely on soundness, zero-knowledge, or simulation

properties; it follows solely from the fact that the public statement exposes $\mathbf{t}$, which is a deterministic function of $(\text{nonce}, \text{vk}_R)$. Hence the adversary distinguishes the challenged instance with certainty, and the nonce-blindness advantage is maximal (equal to 1 under our normalization).

□

Our attack procedure is formally described in Algorithm 1.

Algorithm 1 Adversary $\mathcal{A}$ breaking nonce-blindness

Require: Issuance transcripts $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_R)$ for $i \in \{0, 1\}$, challenge signatures $(\mu_0, \sigma_0), (\mu_1, \sigma_1)$.

- 1: **for** $i \leftarrow 0$ to 1 **do**
- 2: $\mathbf{t}_i \leftarrow \mathcal{H}(\text{nonce}_i \| \text{vk}_R)$
- 3: Parse σ_i and extract the public statement value $\mathbf{t}_i^{(\text{sig})}$
- 4: **end for**
- 5: **if** $\mathbf{t}_0^{(\text{sig})} = \mathbf{t}_0$ **then**
- 6: **return** 0
- 7: **else**
- 8: **return** 1
- 9: **end if**

Theorem 5. *In the random oracle model, the ZCH.LNIBS construction does not satisfy receiver blindness. More precisely, there exists an adversary $\mathcal{A}$ such that*

$$\text{Adv}_{\text{ZCH.LNIBS}}^{\text{RBLD}}(\mathcal{A}) = \left| \Pr \left[\text{Exp}_{\text{ZCH.LNIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda) = 1 \right] - \frac{1}{2} \right| = 1,$$

and $\mathcal{A}$ proceeds as follows.

- Setup. Obtain two receivers $\text{vk}_{R_0}, \text{vk}_{R_1}$ and the corresponding issue transcripts $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_{R_i})$ for $i \in \{0, 1\}$. For each i , the adversary computes $\mathbf{t}_i \leftarrow \mathcal{H}(\text{nonce}_i \| \text{vk}_{R_i})$.
- Challenge. Receive the two outputs (μ_b, σ_b) and $(\mu_{1-b}, \sigma_{1-b})$. From σ_b , extract the public statement $\mathbf{t}_b^{(\text{sig})}$.
- Decision. Output the index i such that $\mathbf{t}_b^{(\text{sig})} = \mathbf{t}_i$. This ensures that $\Pr[\text{Exp}_{\text{ZCH.LNIBS}}^{\text{RBLD}}(\mathcal{A}, 1^\lambda) = 1] = 1$.

Proof. The signature σ contains a publicly readable NIZK proof $\pi = (\pi_{\text{LTC}}, \pi_{\text{VRF}})$. In the NIZK proof π_{LTC} , the statement $\text{stm}_{\text{LTC}} = (\mathbf{t}, \mathbf{A})$ is publicly available. The NIZK proof system is complete, that is, any honestly generated proof accepted by the verifier certifies the asserted relations. The hash function $\mathcal{H}$ is modelled as a random oracle, so collisions in $\mathcal{H}$ occur only with negligible probability. The adversary receives, during issuance, the triples $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_{R_i})$ for $i \in \{0, 1\}$.

The adversary computes $\mathbf{t}_i := \mathcal{H}(\text{nonce}_i \| \text{vk}_{R_i})$ for $i = 0, 1$. These values are deterministic and available to the adversary because the nonces and verification keys appear in the issuance transcripts. When the challenger supplies the two challenge

signatures (μ_b, σ_b) and $(\mu_{1-b}, \sigma_{1-b})$, the adversary parses each σ_i and reads the public statement $\mathbf{t}_i^{(\text{sig})}$ embedded in the NIZK. Parsing here means extracting the statement component from the published signature structure; by construction, this is a public, syntactic operation.

By NIZK completeness, any accepted signature σ contains a statement $\mathbf{t}^{(\text{sig})}$ such that there exists a witness satisfying the relation used in the proof. In particular, for the ZCH construction, the relation includes the equality $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$. Hence, for any valid (accepted) challenge signature σ_i the public statement $\mathbf{t}_i^{(\text{sig})}$ must equal $\mathcal{H}(\text{nonce}_j \parallel \text{vk}_{R_j})$ for the nonce nonce_j and key vk_{R_j} that were used in the issuance witness. Since the adversary previously obtained exactly the two issuance transcripts with nonces $\text{nonce}_0, \text{nonce}_1$ and keys $\text{vk}_{R_0}, \text{vk}_{R_1}$, the value $\mathbf{t}_i^{(\text{sig})}$ must equal either $\mathbf{t}_0$ or $\mathbf{t}_1$.

The only potential failure modes are (i) a hash collision $\mathcal{H}(\text{nonce}_0 \parallel \text{vk}_{R_0}) = \mathcal{H}(\text{nonce}_1 \parallel \text{vk}_{R_1})$, or (ii) an accepted signature that nevertheless does not satisfy the asserted relation (a violation of NIZK completeness). In the ROM, the collision probability is negligible in the security parameter. Completeness of the NIZK guarantees that (ii) happens with negligible probability when proofs are honestly generated. Therefore the probability that $\mathbf{t}_b^{(\text{sig})}$ fails to match the corresponding $\mathbf{t}_i$ is negligible.

Consequently, the adversary recovers the challenge index b with probability at least $1 - \text{negl}(\lambda)$. Formally,

$$\begin{aligned} \Pr[\mathcal{A} \text{ outputs the correct } b] &\geq \\ 1 - \Pr[\text{hash collision}] - \Pr[\text{NIZK completeness failure}] &= 1 - \text{negl}(\lambda). \end{aligned}$$

Thus, the receiver-blindness advantage of $\mathcal{A}$ is overwhelming (equal to 1 up to negligible terms). This conclusion relies only on public exposure of $\mathbf{t}$ in the statement and on the deterministic relation $\mathbf{t} = \mathcal{H}(\text{nonce} \parallel \text{vk}_R)$; it does not require breaking NIZK soundness or VRF security. $\square$

We outline our attack in Algorithm 2.

Algorithm 2 Adversary $\mathcal{A}$ breaking receiver blindness

Require: Issuance transcripts $(\text{psig}_i = \mathbf{x}_i, \text{nonce}_i, \text{vk}_{R_i})$ for $i \in \{0, 1\}$; challenge signatures (μ_b, σ_b) and $(\mu_{1-b}, \sigma_{1-b})$

Ensure: Index b' predicting the hidden bit b

```

1: for  $i \leftarrow 0$  to 1 do
2:    $\mathbf{t}_i \leftarrow \mathcal{H}(\text{nonce}_i \parallel \text{vk}_{R_i})$ 
3: end for
4: Parse  $\sigma_b$  and extract  $\mathbf{t}_b^{(\text{sig})}$ 
5: Parse  $\sigma_{1-b}$  and extract  $\mathbf{t}_{1-b}^{(\text{sig})}$ 
6: if  $\mathbf{t}_b^{(\text{sig})} = \mathbf{t}_0$  or  $\mathbf{t}_{1-b}^{(\text{sig})} = \mathbf{t}_0$  then
7:   return  $b' = 0$ 
8: else
9:   return  $b' = 1$ 
10: end if

```

4.1 Gap in One-More Unforgeability Proof.

In this subsection, we look at the OMUF security of the ZCH.LNIBS construction. We find a gap in their proof that weakens the claimed guarantees. First, we give an informal version of their theorem. Next, we outline their proof idea. Finally, we explain the gap that affects the soundness of the argument.

Informal Theorem: OMUF Security of ZCH.LNIBS. [11] *Suppose the NIZK proof system used in the construction is sound, the VRF has uniqueness, and the hash function H is modeled as a random oracle. Then, if an efficient adversary can query the signer at most l times but still produce $l+1$ valid signatures with non-negligible probability, one can build a reduction that, with essentially the same resources, solves the underlying SIS problem with related probability.*

Proof sketch. The ZCH.LNIBS construction claims OMUF security under the hardness of the SIS problem. The proof follows the standard game-based approach. Starting from the original OMUF game, the reduction first rules out hash collisions in the random oracle, and then uses the uniqueness of the VRF to prevent multiple valid VRF outputs on the same input. In the final step, the adversary’s output of $l+1$ valid message–signature pairs is used to extract a witness from at least one forged signature by relying on the soundness of the underlying NIZK. This extracted witness is then used to solve an SIS instance, thereby completing the reduction.

Gap in the Proof. The extraction step is where the argument becomes problematic. The reduction assumes that the NIZK employed in **Obtain** supports straight-line extraction of the witness from a signature. However, the scheme does not implement any straight-line extractable commitment mechanism (such as *encryption-on-the-sky*). Instead, extraction from a simulation-sound NIZK typically requires rewinding the adversary, which is incompatible with the one-more setting: rewinding risks altering the adversary’s behavior and invalidating the guarantee that exactly $l+1$ distinct signatures can still be obtained. This is in contrast to standard unforgeability, where rewinding only needs to recover a single forgery; in the one-more setting, preserving the adversary’s full output distribution is essential. Consequently, the claimed SIS reduction does not go through as written, since the extractor cannot be applied in the manner required.

5 A Correction to the ZCH.LNIBS Construction

So far, we observe that ZCH.LNIBS does not satisfy the properties of *blindness*. It also has a gap in the OMUF security proof. We fix both things in a step-by-step manner. *Sketch of the idea.* Earlier, the expression about the presignature is $\mathbf{Ax} = \mathbf{t}$; where $\|\mathbf{x}\| \leq \beta$. In the **Obtain** algorithm, the receiver prepares a NIZK proof for it. It publishes $\mathbf{t}$ as a statement. This creates a problem in the blindness security. To handle this situation, the designer needs to rewrite the expression as

$$\mathbf{Ax} + \mathbf{Rr} = \mathbf{Bs} + \mathcal{H}(\mu_0). \quad (1)$$

In this expression, the user will compute the right-hand side, and the issuer will compute the left-hand side. At the end, the receiver provides an NIZK proof for the

witness $\mathbf{x}, \mathbf{r}, \mathbf{s}$, and corresponding statements are $(\mu_0, \mathbf{A}, \mathbf{R}, \mathbf{B})$. The same expression is also used in the BCZY.LNIBS construction [2].

Consider the blindness game, in the final signature, $(\mu_0, \mathbf{A}, \mathbf{R}, \mathbf{B})$ are present but they reveal no information about the receiver public key, the presignature, and the nonce pair. Note that the signer is computing $\mathbf{Ax} + \mathbf{Rr} = \mathbf{t}$, where $\mathbf{t} = \mathbf{Bs} + \mathcal{H}(\mu_0)$ is a fixed target vector for a fixed receiver public key. Now consider the unforgeability proof. A malicious receiver has to produce a new pair $(\mathbf{x}^*, \mathbf{r}^*)$ for the expression $\mathbf{Ax} + \mathbf{Rr} = \mathbf{t}$. This is an instance of the rOM-ISIS assumption, which is proposed in the Baldimtsi et al. work [2]. This shows that our approach is similar to BCZY.LNIBS approach, and the security of our construction also relies on the rOM-ISIS assumption.

1. *Modifications in the Receiver's Key Generation Algorithm.* In ZCH.LNIBS, the receiver samples a short vector $\mathbf{s}$ and computes $\mathbf{Bs}$. The receiver keeps $\mathbf{s}$ as the secret key and publishes $\mathbf{Bs}$ as the verification key. However, this design allows a malicious signer to link a message–signature pair with its corresponding presignature–nonce pair. To prevent this linkage, we slightly modify the key-generation phase: first, uniformly sample a random bit string μ_0 , then compute $\mathbf{t} \leftarrow \mathcal{H}(\mu_0) + \mathbf{Bs}$, where $\mathbf{s}$ is a short vector. The verification key of the receiver is now $\mathbf{t}$, while the secret key remains $(\mathbf{s}, \mu_0)$.

2. *Modifications in Issue algorithm.* Previously, the issuer directly signed the input vector, which introduced a vulnerability. A natural fix is to add randomness before signing. Concretely, the issuer samples a random vector $\mathbf{r}^* \in_U \mathcal{R}_q$ and adds it to $\mathbf{t}$. Then it runs **SamplePre** on $\mathbf{t} - \mathbf{r}^*$ to compute $\mathbf{x}$. Finally, it outputs $\mathbf{r}^*$ as the nonce and $\mathbf{x}$ as the presignature. This modification restores the blindness property. However, it breaks unforgeability: an adversary can simply choose $\mathbf{x}$ at random, compute $\mathbf{A} \cdot \mathbf{x}$, and set $\mathbf{t} - \mathbf{Ax}$ as the nonce.

Instead of sampling only a vector, one can use a lattice-based trapdoor. Let $\mathbf{R} \in \mathcal{R}_q^{n_1 \times m_1}$ be a random matrix. The signer first samples a short vector $\mathbf{r}$ and computes $\mathbf{t} - \mathbf{R} \cdot \mathbf{r}$. Then it runs **SamplePre** on this value to obtain the presignature $\mathbf{x}$. The signer outputs $\mathbf{r}$ as the nonce and $\mathbf{x}$ as the presignature. In the unforgeability game, the adversary receives l pairs $(\mathbf{x}, \mathbf{r})$ and must produce a new pair $(\mathbf{x}_1, \mathbf{r}_1)$ such that $\mathbf{t} = \mathbf{Ax}_1 + \mathbf{Rr}_1$. Due to this expression, the security of the construction relies on the non-standard rOM-ISIS assumption.

The same trick of introducing randomness via an auxiliary matrix $\mathbf{R}$, and a short random vector $\mathbf{r}$ was also used in the construction of Baldimtsi *et al.* They sampled a vector $\mathbf{r} \in \{-1, +1\}^{2m}$, and a random vector $\mathbf{R}$; and then compute the $\mathbf{Rr}^4$.

3. *Modifications in the Obtain Algorithm.* Clearly, the receiver first verifies whether the presignature and nonce are valid or not by checking the Equation: 1. Further, it provides the well-formedness proof for this expression. In this NIZK proof, it keeps $(\mathbf{x}, \mathbf{s}, \mathbf{r})$ as witness; and $\mathbf{A}, \mathbf{B}, \mathbf{R}, \mu_1$ as statement. In the ZCH.LNIBS construction, it only performs a VRF evaluation and its NIZK proof. The output of the VRF is treated as a message. For the OMUF security proof, the simulator needs to extract the witness. To do that, the receiver may use some witness extractable constructions. The most popular technique is the *encryption-to-the-sky*. Here, the receiver encrypts witness $(\mathbf{x}, \mathbf{s}, \mathbf{r})$ using a lattice-based public-key encryption. The next task for the

⁴ For more details see Appendix B, Figure 7.

receiver is to compute the message. We propose using VRF on $(\mathbf{x}, \mathbf{r})$, and provide a NIZK proof of this computation to ensure the reusability property.

We present the algorithm in Figure 6. Here, the parameters $n_1, m_1, q, \beta_0, \beta_1, \beta_2$ are functions of security parameters λ . In particular, $n = \text{poly}(\lambda)$, $m \geq \mathcal{O}(n \log q)$, which are chosen in such a way that the underlying rOM-ISIS problem should be hard. Similarly, we need to identify efficient parameters for our secure Tagged-based version of ZCH.LNIBS. Since our construction closely follows the Baldimtsi et al. construction [2], we do not present the security proof.

6 Conclusion

In this work, we revisited the lattice-based blind signature from the standard assumption. This construction is proposed by Zhang *et al.* [11]. We prove that the construction does not satisfy the fundamental properties of a blind signature; that is, it does not offer blindness. Also, the unforgeability proof has a flaw. We suggested a modification to secure this construction. However, our construction is similar to the construction of Baldimtsi *et al.* [2]. So the security of our proposal relies on the random-one-more-ISIS assumption. Therefore, designing a lattice-based construction from standard assumptions is still open and an interesting research direction.

A Basic Cryptographic Primitives

A.1 Non-interactive Zero Knowledge Proof.

A non-interactive zero-knowledge (NIZK) proof system Π for an NP-language $\mathcal{L}$ consists of a tuple of probabilistic polynomial-time (PPT) algorithms (NIZK.Setup, NIZK.Proof, NIZK.Verify). The NIZK.Setup algorithm outputs a common reference string crs . Given crs , a statement stm , and a witness wtn , the prover runs NIZK.Proof($\text{crs}, \text{stm}, \text{wtn}$) to generate a proof π for the language $\mathcal{L}$. The verifier, on input $(\text{crs}, \text{stm}, \pi)$, runs NIZK.Verify($\text{crs}, \text{stm}, \pi$) to accept or reject. The proof system must satisfy the standard properties of completeness, soundness, and zero-knowledge.

- $\text{crs} \leftarrow \text{NIZK.Setup}(1^\lambda)$. It takes security parameter λ as input and outputs a common reference string crs .
- $\pi \leftarrow \text{NIZK.Proof}(\text{crs}, \text{stm}, \text{wtn})$. It generates a proof π on input a statement stm , and a witness wtn .
- $b \leftarrow \text{NIZK.Verify}(\text{crs}, \text{stm}, \pi)$ The verifier algorithm outputs 1 if π is accepted as a valid proof, and 0 otherwise.

Security Properties.

Definition 8 (Completeness). A NIZK proof system Π is complete, if all security parameter $\lambda \in \mathbb{N}$, $\text{crs} \leftarrow \text{NIZK.Setup}(1^\lambda)$ and $(\text{wtn}, \text{stm}) \in \mathcal{L}$, the following condition should holds.

$$\Pr[\text{NIZK.Verify}(\text{crs}, \text{stm}, \pi) = 1 \mid \pi \leftarrow \text{NIZK.Proof}(\text{crs}, \text{stm}, \text{wtn})] = 1.$$

Lattice-based NIBS Construction**Statement / Witness Relation.**

$\text{stm} \leftarrow (\text{crs}, \mathbf{R}, \mathbf{A}, \mathbf{B}, \text{pke.pk}, \text{ct}, \mu_0, \mu_1)$ and $\text{wtn} \leftarrow (\mathbf{x}, \mathbf{r}, \mathbf{s}, b, r)$.

Membership: wtn is a valid witness for the statement stm if:

1. $\|\mathbf{x}\| \leq \sqrt{2}\beta_0$, $\|\mathbf{s}\| \leq \sqrt{2}\beta_1$, $\mathbf{r} \in \sqrt{2}\beta_2$;
2. $\mathbf{Ax} + \mathbf{Rr} = \mathbf{B} \cdot \mathbf{s} + \mathcal{H}(\mu_0)$; $\mathbf{A}, \mathbf{B}, \mathbf{R} \in \mathcal{R}_q^{n_1 \times m_1}$.
3. $\text{ct} = \text{PKE.Enc}(\text{pke.pk}, \mathbf{x} \parallel \mathbf{s} \parallel \mathbf{r}; r)$
4. $\mu_1 \leftarrow \text{VRF.Eval}(\mathbf{x}, \mathbf{s})$.

Key Generation.

Issuer	Receiver
<u>KeyGen(1^λ)</u>	<u>RecKeyGen(1^λ)</u>
1. $\text{params} \leftarrow \text{SetUp}(1^\lambda)$. 2. $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \leftarrow \text{TrapGen}(\text{params})$. 3. $\text{sk}_\mathbf{I} \leftarrow \mathbf{T}_\mathbf{A}$; $\text{vk}_\mathbf{I} \leftarrow \mathbf{A}$; $\mathbf{A} \in \mathcal{R}_q^{n_1 \times m_1}$. 4. return $(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{I})$.	1. Sample a small vector $\mathbf{s} \in \mathcal{R}_q^{m_1}$ with $\ \mathbf{s}\ \leq \beta_1$ 2. Sample $\mu_0 \xleftarrow{\\$} \{0, 1\}^\lambda$. 3. Sample a bit $b \xleftarrow{\\$} \{0, 1\}$. 4. Compute $\mathbf{t} \leftarrow \mathbf{B} \cdot \mathbf{s} + \mathcal{H}(\mu_0)$; $\mathbf{B} \in \mathcal{R}_q^{n_1 \times m_1}$. 5. Set $\text{sk}_\mathbf{R} \leftarrow (\mathbf{s}, \mu_0)$, $\text{vk}_\mathbf{R} \leftarrow \mathbf{t}$. 6. return $(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{R})$.

Signature Generation.

<u>$(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{R})$</u>	<u>$(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{I}, \text{psig}, \text{nonce})$</u>
1. Sample a nonce: a small vector $\ \mathbf{r}\ \leq \beta_2$; $\mathbf{r} \in \mathcal{R}_q^{m_1}$. 2. Compute $\mathbf{x} \xleftarrow{\\$} \text{SamplePre}(\mathbf{A}, \mathbf{T}_\mathbf{A}, \mathbf{t} - \mathbf{Rr}, \beta_0)$. 3. Set $\text{psig} \leftarrow \mathbf{x}$, $\text{nonce} \leftarrow \mathbf{r}$. 4. return $(\text{psig}, \text{nonce})$.	1. Parse $\text{sk}_\mathbf{R} = (\mathbf{s}, b, \mu_0)$. 2. Given $\text{vk}_\mathbf{R}, \mathbf{x}, \mathbf{r}$, check whether: $\mathbf{Ax} + \mathbf{Rr} \stackrel{?}{=} \mathbf{B} \cdot (\mathbf{s}, b)^\top + \mathcal{H}(\mu_0),$ $\ \mathbf{x}\ \leq \sqrt{2}\beta_0, \quad \ \mathbf{r}\ \leq \sqrt{2}\beta_1.$ 3. Check fails, output $\perp$ and abort. 4. Otherwise, sample $r \xleftarrow{\\$} \{0, 1\}^\lambda$ and compute $\text{ct} \leftarrow \text{PKE.Enc}(\text{pke.pk}, \mathbf{s} \parallel \mathbf{r} \parallel \mathbf{x} \parallel b; r)$. 5. Compute $\mu_1 \leftarrow \text{VRF.Eval}(\mathbf{x}, \mathbf{s})$. 6. Generate NIZK proof π: – Step-2 relation – Proof of PKE – VRF evaluation 7. Output $\mu \leftarrow (\mu_0, \mu_1)$ and $\sigma \leftarrow (\pi, \text{ct})$.

Verification.

$\text{Verify}(\text{vk}, \mu, \sigma) \rightarrow \{0, 1\}$

1. Parse $\mu = (\mu_0, \mu_1)$ and $\sigma = (\pi, \text{ct})$.
2. Output 1 if $\text{NIZK.Verify}(\text{stm}, \pi) = 1$. else 0.

Fig. 6. Our Proposal: Lattice-based non-interactive blind signature scheme.

Definition 9 (Soundness). A NIZK proof system Π for a language $\mathcal{L}$ is sound if for every PPT adversary $\mathcal{A}$, the probability that $\mathcal{A}$ produces a statement $\text{stm} \notin \mathcal{L}$ along with a proof π such that $\text{NIZK.Verify}(\text{crs}, \text{stm}, \pi) = 1$ is negligible in the security parameter λ . Formally,

$$\Pr[\text{NIZK.Verify}(\text{crs}, \text{stm}, \pi) = 1 \wedge \text{stm} \notin \mathcal{L} \mid (\text{stm}, \pi) \leftarrow \mathcal{A}(\text{crs}, 1^\lambda)] \leq \text{negl}(\lambda).$$

Definition 10 (Honest-Verifier Zero-Knowledge (HVZK)). A NIZK proof system Π for an NP-language $\mathcal{L}$ is honest-verifier zero-knowledge if there exists a PPT simulator Simu such that for every $(\text{stm}, \text{wtn}) \in \mathcal{L}$, the distribution of simulated proofs $\pi_{\text{Simu}} \leftarrow \text{Simu}(\text{stm})$ is computationally indistinguishable from the distribution of real proofs $\pi_{\text{Real}} \leftarrow P(\text{stm}, \text{wtn})$. Formally, for every PPT distinguisher $\mathcal{D}$,

$$\left| \Pr[\mathcal{D}(\text{stm}, \pi_{\text{Real}}) = 1 \mid \pi_{\text{Real}} \leftarrow \text{NIZK.Proof}(\text{crs}, \text{stm}, \text{wtn})] - \Pr[\mathcal{D}(\text{stm}, \pi_{\text{Simu}}) = 1 \mid \pi_{\text{Simu}} \leftarrow \text{Simu}(\text{stm})] \right| \leq \text{negl}(\lambda).$$

A.2 Verifiable Random Function.

A verifiable random function (VRF) consists of five different polynomial-time algorithms $\text{VRF} = \{\text{VRF.SetUp}, \text{VRF.KeyGen}, \text{VRF.Eval}, \text{VRF.Verify}\}$.

- $\text{params} \leftarrow \text{VRF.SetUp}(1^\lambda)$. It generates the required parameters for the VRF.
- $(\text{sk}_{\text{VRF}}, \text{vk}_{\text{VRF}}) \leftarrow \text{VRF.KeyGen}(\text{params})$. It takes params as input, and outputs the key pair $(\text{sk}_{\text{VRF}}, \text{vk}_{\text{VRF}})$ for VRF.
- $(R, \pi_{\text{VRF}}) \leftarrow \text{VRF.Eval}(E, \text{sk}_{\text{VRF}})$. It takes an input E , and the secret key sk_{VRF} , and it outputs the VRF value R and the corresponding NIZK proof π_{VRF} .
- $0/1 \leftarrow \text{VRF.Verify}(E, R, \pi_{\text{VRF}}, \text{vk}_{\text{VRF}})$. The verifier verifies the proof π_{VRF} using E, R and the verification key vk_{VRF} .

Properties of VRF. The VRF has three properties: *correctness*, *pseudorandomness*, *uniqueness*.

– **Correctness.** Let $(\text{sk}_{\text{VRF}}, \text{vk}_{\text{VRF}}) \leftarrow \text{VRF.KeyGen}(\text{params})$ be a key pair honestly generated from the public parameters params . For any input E , let $(R, \pi_{\text{VRF}}) \leftarrow \text{VRF.Eval}(\text{sk}_{\text{VRF}}, E)$. Then, $\text{VRF.Verify}(E, R, \pi_{\text{VRF}}, \text{vk}_{\text{VRF}}) = 1$ holds with probability 1.

– **Pseudorandomness.** We define the *pseudorandomness* of a VRF through the following distinguishing game.

An adversary $\mathcal{A}$ is given access to either a uniformly random bit string $R \xleftarrow{\$} \{0, 1\}^{\ell(\lambda)}$ (where $\ell(\lambda)$ is the output length of the VRF), or to the output R of the evaluation algorithm $\text{VRF.Eval}(E, \text{sk}_{\text{VRF}})$, where sk_{VRF} is generated using $\text{VRF.KeyGen}(\text{params})$. The adversary's goal is to distinguish between these two cases. The distinguishing advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{\text{VRF}}^{\text{RND}}(\mathcal{A}) = \left| \Pr_{R \xleftarrow{\$} \{0, 1\}^{\ell}}[\mathcal{A}(R) = 1] - \Pr_{\substack{(\text{sk}_{\text{VRF}}, \text{vk}_{\text{VRF}}) \leftarrow \text{VRF.KeyGen}(\text{params}) \\ (R, \pi_{\text{VRF}}) \leftarrow \text{VRF.Eval}(E, \text{sk}_{\text{VRF}})}}[\mathcal{A}(R) = 1] \right|.$$

We say that the VRF satisfies *pseudorandomness* if $\text{Adv}_{\text{VRF}}^{\text{RND}}(\mathcal{A})$ is negligible in the security parameter λ for all probabilistic polynomial-time adversaries $\mathcal{A}$.

– **Uniqueness.** We say that a VRF satisfies *uniqueness* if, for the same input E , there does not exist a value $R^* \neq R$ such that

$$\text{VRF.Verify}(E, R, \pi_{\text{VRF}}, \text{vk}_{\text{VRF}}) = \text{VRF.Verify}(E, R^*, \pi_{\text{VRF}}^*, \text{vk}_{\text{VRF}}) = 1.$$

B Lattice-based Blind Signature.

We present the BCGY NIBS construction [2,1] in Figure 7. For a more detailed idea about the parameters of Figure 7, we refer to Section 8 of the paper [1].

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Lattice-based NIBS Construction	
Statement / Witness Relation. $\text{stm} \leftarrow (\text{crs}, \mathbf{R}, \mathbf{A}, \mathbf{B}, \text{pke.pk}, \text{ct}, \mathbf{w}, \mu_0)$ and $\text{wtn} \leftarrow (\mathbf{x}, \mathbf{r}, \mathbf{s}, b, r)$. Language: wtn is a valid witness for the statement stm if the following hold: 1. $\ \mathbf{x}\ \leq \sqrt{2}\beta$, $\ \mathbf{s}\ \leq 1 + 2^{(\lambda/2-1)} \cdot \beta/m$, $\mathbf{r} \in \{\pm 1\}^{2m}$; 2. $\mathbf{Ax} + \mathbf{Rr} = \mathbf{B} \cdot (\mathbf{x}, b)^\top + \mathcal{H}(\mu_0)$. 3. $\text{ct} = \text{PKE.Enc}(\text{pke.pk}, \mathbf{x} \parallel \mathbf{s} \parallel \mathbf{r} \parallel b; r)$ 4. $(1-2b) \cdot \mathbf{w} = \mathbf{z}^\perp - \mathbf{x}^\perp$, and $\theta(1-\theta) = 0$.	
Key Generation.	
Issuer $\text{KeyGen}(1^\lambda)$ 1. $\text{params} \leftarrow \text{SetUp}(1^\lambda)$. 2. $(\mathbf{A}, \mathbf{T}_\mathbf{A}) \leftarrow \text{TrapGen}(\text{params})$. 3. $\text{sk}_\mathbf{I} \leftarrow \mathbf{T}_\mathbf{A}$; $\text{vk}_\mathbf{I} \leftarrow \mathbf{A}$; $\mathbf{A} \in \mathcal{R}_q^{n \times (2m+1)}$. 4. return $(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{I})$.	Receiver $\text{RecKeyGen}(1^\lambda)$ 1. Sample the two halves of $\mathbf{s} \in \mathcal{R}_q^{2m}$ as: $\mathbf{s}^\top \xleftarrow{\\$} D_{\mathbf{z}_q, \rho/m}^m$, $\mathbf{s}^\perp \xleftarrow{\\$} \left[\frac{-2^{(\lambda/2-1)}}{m\sqrt{m}}\beta, \frac{2^{(\lambda/2-1)}}{m\sqrt{m}}\beta \right]^m$. 2. Sample $\mu_0 \xleftarrow{\\$} \{0, 1\}^\lambda$. 3. Sample a bit $b \xleftarrow{\\$} \{0, 1\}$. 4. Compute $\mathbf{t} \leftarrow \mathbf{B} \cdot (\mathbf{s}, b)^\top + \mathcal{H}(\mu_0)$. 5. Set $\text{sk}_\mathbf{R} \leftarrow (\mathbf{s}, \mu_0, b)$, $\text{vk}_\mathbf{R} \leftarrow \mathbf{t}$. 6. return $(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{R})$.
Signature Generation.	
$(\text{psig}, \text{nonce}) \leftarrow \text{Issue}(\text{sk}_\mathbf{I}, \text{vk}_\mathbf{R})$ 1. Sample a nonce $\mathbf{r} \in \{-1, +1\}^{2m}$. 2. Compute $\mathbf{x} \xleftarrow{\\$} \text{SamplePre}(\mathbf{A}, \mathbf{T}_\mathbf{A}, \mathbf{t} - \mathbf{Rr}, \beta)$. 3. Set $\text{psig} \leftarrow \mathbf{x}$, $\text{nonce} \leftarrow \mathbf{r}$. 4. return $(\text{psig}, \text{nonce})$.	$(\mu, \sigma) \leftarrow \text{Obtain}(\text{sk}_\mathbf{R}, \text{vk}_\mathbf{I}, \text{psig}, \text{nonce})$ 1. Parse $\text{sk}_\mathbf{R} = (\mathbf{s}, b, \mu_0)$. 2. Given $\text{vk}_\mathbf{R}, \mathbf{x}, \mathbf{r}$, check whether: $\mathbf{Ax} + \mathbf{Rr} \stackrel{?}{=} \mathbf{B} \cdot (\mathbf{s}, b)^\top + \mathcal{H}(\mu_0),$ $\ \mathbf{x}\ \leq \sqrt{2}\beta, \quad \mathbf{r} \in \{\pm 1\}^{2m}.$ 3. Check fails, output $\perp$ and abort. 4. Otherwise, sample $r \xleftarrow{\\$} \{0, 1\}^\lambda$ and compute $\text{ct} \leftarrow \text{PKE.Enc}(\text{pke.pk}, \mathbf{s} \parallel \mathbf{r} \parallel \mathbf{x} \parallel b; r)$. 5. Compute $\mathbf{w} \leftarrow (1-2b)(\mathbf{x}^\perp - \mathbf{s}^\perp)$. 6. Generate NIZK proof π: – Step-2 relation – Proof of PKE – $\mathbf{w} \leftarrow (1-2b)(\mathbf{x}^\perp - \mathbf{s}^\perp)$ 7. Output $\mu \leftarrow (\mathbf{w}, \mu_0)$ and $\sigma \leftarrow (\pi, \text{ct})$.
Verification.	
$\text{Verify}(\text{vk}, \mu, \sigma) \rightarrow \{0, 1\}$ 1. Parse $\mu = (\mathbf{w}, \mu_0)$ and $\sigma = (\pi, \text{ct})$. 2. Output 1 if $\text{NIZK.Verify}(\text{stm}, \pi) = 1$. else 0.	

Fig. 7. Lattice-based non-interactive blind signature scheme.